

AMATH 732

Quiz 3; 30 Minutes

Question 1: 5 marks Find the approximate roots of

$$p(x) = x^3 - x^2 + \epsilon$$

to first order in $\epsilon \ll 1$.

Question 1: 5 marks Consider the polynomial equation

$$p(x) = x^2 + (\epsilon^2 - \epsilon - 1)x - \epsilon^3 - 2\epsilon^2 - \epsilon - 2 = 0$$

Use regular perturbation theory to find the solution to second order.